

B.Sc. 3rd Semester (Honours) Examination, 2023-24

COMPUTER SCIENCE

Course ID : 31513

Course Code : SH-CSC-303/C-7

Course Title : Computer Networks

[Syllabus 2017-18]

Time : 1 Hour 15 Minutes

Full Marks : 25

The figures in the right margin indicate marks.

Candidates are required to answer in their own words as far as practicable.

UNIT-I

1. Answer *any five* from the following questions:

5×1=5

- a) Define Modem.
- b) What is token ring?
- c) What is Round Trip Time?
- d) What is congestion?
- e) What is TDM?
- f) What is HUB?
- g) What is piggy backing?
- h) Define data rate.

UNIT-II

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2. Answer *any two* from the following questions:

2×5=10

- a) Differentiate between Router and Switch.
- b) Write short note on UDP.
- c) Describe different network topologies.
- d) Explain in brief: stop and wait ARQ.

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UNIT-III

3. Answer *any one* from the following questions:

1×10=10

- a) Describe the process of three way handshake
Write short note on Ethernet LAN. 5+5
- b) Describe OSI reference model. 10

B.Sc. 3rd Semester (Honours) Examination, 2023-24

COMPUTER SCIENCE

Course ID : 31513

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Course Title : Communication and Computer Networks

[Syllabus 2022-23]

Time : 1 Hour 15 Minutes

Full Marks : 25

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UNIT-I

Answer *any five* from the following questions:

5×1=5

- a) Define Modem.
- b) What is token ring?
- c) What is Round Trip Time?
- d) What is congestion?
- e) What is TDM?
- f) What is HUB?
- g) What is piggy backing?
- h) Define flooding.

[Turn Over]

UNIT-II

7. Answer *any two* from the following questions:

2×5=10

- a) Differentiate between Router and Switch.
- b) Write short note on UDP.
- c) Describe different network topologies.
- d) Explain in brief: stop and wait ARQ.

UNIT-III

8. Answer *any one* from the following questions:

1×10=10

- a) Describe OSI reference model. 10
- b) Describe the process of three way handshake.
Write short note on Ethernet LAN. 5+5

BANKURA SAMMILANI COLLEGE

Dept. of Computer Science

INTERNAL ASSESSMENT EXAMINATION – 2023

SEM- 3 (H)

PAPER: C-7

Full marks: 5

Time: 30 min

Answer any five questions:

1. What is internetworking?
- ~~2.~~ Which services are provided by data link layer?
- ~~3.~~ What is bandwidth?
- ~~4.~~ What do you mean by bit rate and baud rate?
- ~~5.~~ Write down advantages and disadvantages of twisted pair cable.
6. Write down the steps to convert an analog signal into a digital data.
- ~~7.~~ Why we use flow control in data communication?
- ~~8.~~ What is piggybacking?

B.Sc. 3rd Semester (Honours) Examination, 2023-24

COMPUTER SCIENCE

Course ID : 31523

Course Code : SH-CSC-303/C-7

**Course Title : Communication and Computer Networks
(Practical)**

[Syllabus 2022-23]

Time : 2 Hours

Full Marks : 15

The figures in the right margin indicate marks.

*Candidates are required to answer in their own
words as far as practicable.*

[Problem = 10, LNB + Viva voce = 05]

1. Answer any **one** of the following questions:

10×1=10

- a) Simulate and implement go back n sliding window protocol.
- ~~b)~~ Simulate and implement distance vector routing algorithm.
- c) Simulate and implement Dijkstra algorithm for shortest path routing.
- d) Simulate and implement stop and wait protocol for noisy channel.
- e) Simulate and implement selective repeat sliding window protocol.
- f) Simulate Cyclic Redundancy Check (CRC) error detection algorithm for noisy channels.

[Turn Over]

B.Sc. 3rd Semester (Honours) Examination, 2023-24

COMPUTER SCIENCE

Course ID : 31523

Course Code : SH-CSC-303/C-7

Course Title : Computer Networks (Practical)

[Syllabus 2017-18]

Time : 2 Hours

Full Marks : 15

The figures in the right margin indicate marks.

Candidates are required to answer in their own words as far as practicable.

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